

Multiple Choice (10 marks)

1. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
2. Which of the following decimal numbers has an exact representation in binary notation?
 - (a) 0.1
 - (b) 0.2
 - (c) 0.3
 - (d) 0.5
3. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
 - (a) `**`
 - (b) `//`
 - (c) `is`
 - (d) `not in`
4. A large Java program was tested extensively, and no errors were found. What can be concluded?
 - (a) All of the preconditions in the program are correct.
 - (b) All of the postconditions in the program are correct.
 - (c) The program may have bugs.
 - (d) Every method in the program may safely be used in other programs.
5. Which of the following comes closest to being a perfectly secure encryption scheme?
 - (a) The Caesar Cipher, a substitution cipher
 - (b) DES (Data Encryption Standard), a symmetric-key algorithm
 - (c) Enigma, a transposition cipher
 - (d) One-time pad

True / False (10 marks)

Answer True or False

6. True or False: Merge sort has average-case and worst-case running times of $O(n \log n)$, making it efficient for large data sets.
7. True or False: In Python 3, the expression `'abc'[-1]` results in an error due to out-of-bounds indexing.
8. True or False: The regular expression $a^*(c + d) + b(c + d)$ describes the same set of strings as $(a^* + b)^*(c + d)$.
9. True or False: The value of the Python expression `1 + 3 % 3` is 4.
10. True or False: Local caching of files in distributed file systems guarantees that all machines always have the same view of a file.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the roots of a quadratic equation are numerically equal but opposite in sign or not.
12. Write a function to check whether a list contains the given sublist or not.

End of Paper